COMPACT AND NONCOMPACT INVERSE HANKEL CHOICES

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Problem 1 (Peller [2, Chapter 12, §13, p. 537]; Megretskii–Peller–Treil [1, §III.7, p. 295])**.** In the proof of [2, Chapter 12, Theorem 9.1] for a self-adjoint operator Γ with $\operatorname{Ker}\Gamma = \{0\}$ satisfying (C1)–(C3): "it is unknown if it is possible to choose $q \notin \operatorname{Range} R$ so that $I - Q$ is compact. It is also unclear whether it is possible to find q so that $I - Q$ is noncompact".

Setting. We use the notation of [2, Chapter 12, §§9–12]. Let Γ be a bounded self-adjoint operator satisfying (C1)–(C3) with $\operatorname{Ker}\Gamma = \{0\}$. In §12, W is multiplication by t on the direct integral $\mathcal{K} = \int_{\sigma(W)}^{\oplus} E(t)\,\mathrm{d}\rho(t)$, J is the multiplication defined there, $R = JW$, and the first level $\mathcal{K}_0 = \{\varphi(t)e_1\}$ is identified with $L^2(\rho)$. The vector $q_0 \in \mathcal{K}_0$ must satisfy (1) $\|q_0\| = 1$; (2) $q_0(t) \neq 0$, ρ-a.e.; (3) $q_0 \notin \operatorname{Range} W$. Then $q = Jq_0$, $p = Rq = Wq_0$, $A^2 = W^2 - (\cdot, p)p$, and Q is the contraction with $QW^{1/2} = |A|^{1/2}$. Since J is unitary and commutes with W, $q \notin \operatorname{Range} R$ is equivalent to (3). The proof of Lemma 14.2 in [2] uses one further property of q_0: for almost all $t \in \sigma_2^{(\pm)}$,

$$\int_{\sigma(W)\setminus\sigma_2^{(\pm)}} \frac{|p(s)|^2}{|s^2 - t^2|}\,\mathrm{d}\rho(s) < \infty \qquad (p(s) = sq_0(s)). \tag{14.8}$$

This is [2, Chapter 12, (14.8)]; the proof of Lemma 14.2 constructs such a q_0. By (C3), $\sigma_2^{(\pm)}$ lies in the set Δ on which ρ and the scalar spectral measure λ of $|A|$ restricted to $\mathcal{K}_0$ have the same null sets [2, Chapter 12, p. 534]. So "almost all" in Equation (14.8) refers to the same null sets for every admissible q_0. The proof of [2, Chapter 12, Lemma 13.5] uses Equation (14.8) even when $I - Q$ is compact.

Theorem 1. *Let Γ, W, J, R, ρ be as above, and let $q_0^\sharp$ be any function satisfying (1)–(3), for instance the function constructed in the proof of* [2, Chapter 12, Lemma 14.2]*. There are positive functions φ_{c}, φ_{n} on $\sigma(W) \setminus \{0\}$, each taking finitely many values on every compact subset of $(0, \infty)$, such that both $q_0 = \varphi_{\mathrm{c}} q_0^\sharp$ and $q_0 = \varphi_{\mathrm{n}} q_0^\sharp$ satisfy (1)–(3), and:*

(a) *for $q_0 = \varphi_{\mathrm{c}} q_0^\sharp$, the operator $I - Q$ is Hilbert–Schmidt;*
(b) *for $q_0 = \varphi_{\mathrm{n}} q_0^\sharp$, the operator $I - Q$ is noncompact.*

For every $t > 0$, the integral in Equation (14.8) *is finite for q_0 if and only if it is finite for $q_0^\sharp$.*

Thus, for every Γ in [2, Chapter 12, Theorem 9.1] with trivial kernel, both choices exist, and both can be made inside the class of vectors used in the proof of that theorem.

Remark 1. By (C2) the operator W is not invertible. Since $\operatorname{Ker} W = \{0\}$ and isolated points of the spectrum of a self-adjoint operator are eigenvalues, 0 is an accumulation point of $\sigma(W)$ and $\rho((0, s)) > 0$ for all $s > 0$. If W were invertible, then $\operatorname{Range} R = \mathcal{K}$ and no $q \notin \operatorname{Range} R$ would exist. This case is excluded by (C2).

Remark 2. With $q_0 = \varphi_{\mathrm{c}} q_0^\sharp$, the operator $A^* - \mathcal{J}J = Q^*\mathcal{J}QJ - \mathcal{J}J$ is compact, so $\sigma_F(A^*) = \sigma_F(\mathcal{J}J)$. This replaces [2, Chapter 12, Lemmas 13.2 and 13.3], together with the restriction $|\sin\omega_1| \leq \chi$, as Peller indicates on p. 537.

Lemma 1. *$\mathcal{K}_0$ reduces W and $|A|$, and $Q = Q_0 \oplus I$ on $\mathcal{K}_0 \oplus \mathcal{K}_0^\perp$. Here, with M multiplication by s on $L^2(\rho)$,*

$$Q_0 M^{1/2} = |A_0|^{1/2}, \qquad A_0^2 = M^2 - (\cdot, p)p.$$

Proof. By (2), q_0 is cyclic for M, so $\mathcal{K}_0$ is the smallest W-invariant subspace containing $Jq = q_0$. This is [2, Chapter 12, Lemma 10.6] and its proof. □

Date: 13 October 2026.

Fix the scale. Replacing Γ by $\Gamma/\|\Gamma\|$ multiplies W, $|A|$ and p by $1/\|\Gamma\|$, and leaves Q, q_0 and (1)–(3) unchanged. We assume $\|W\| \le 1$. For integers $k \ge 0$ put $I_k = (2^{-k-1}, 2^{-k}]$, and let $k_1 < k_2 < \cdots$ enumerate the k with $\rho(I_k) > 0$. This set is infinite by the first remark, and $k_l \ge l - 1$. Every q_0 satisfying (1) and (2) determines the probability measure $\mu = |q_0|^2\rho$, which has the same null sets as ρ.

Lemma 2. *Suppose that q_0 satisfies (1) and (2), and that $m_l = \mu(I_{k_l})$ satisfy, for some $\beta \ge 1$ and all $l, n \ge 1$,*

$$m_l \le \frac{\beta}{l^2}, \qquad \sum_{l>n} m_l \ge \frac{1}{\beta(n+1)}.$$

Then $q_0 \notin \operatorname{Range} W$, and $I - Q_0$ is a Hilbert–Schmidt operator on $L^2(\rho)$.

Proof. Since $\operatorname{Range} M = \{f : f(s)/s \in L^2(\rho)\}$,

$$\int \frac{|q_0(s)|^2}{s^2}\,\mathrm{d}\rho(s) \ge \sum_{l>n} 4^{k_l} m_l \ge \frac{4^n}{\beta(n+1)} \qquad (n \ge 1),$$

which is unbounded in n. For $t > 0$ put $g_t(s) = sq_0(s)/(s^2+t)$, so $g_t = (M^2+t)^{-1}p$, and

$$d(t) = 1 - ((M^2+t)^{-1}p, p) = \int \frac{t}{s^2+t}\,\mathrm{d}\mu(s) > 0.$$

Direct multiplication gives the rank-one resolvent formula $(A_0^2+t)^{-1} - (M^2+t)^{-1} = (\cdot, g_t)g_t/d(t)$. For a positive operator X, $X^{1/4} = c\int_0^\infty t^{1/4}(t^{-1} - (X+t)^{-1})\,\mathrm{d}t$ with $c = \sin(\pi/4)/\pi$, the integral converging in norm. Subtracting the formulas for $X = M^2$ and $X = A_0^2$, we obtain

$$M^{1/2} - |A_0|^{1/2} = c\int_0^\infty \frac{t^{1/4}}{d(t)}(\cdot, g_t)g_t\,\mathrm{d}t,$$

and $M^{1/2} - |A_0|^{1/2} = (I - Q_0)M^{1/2}$. Put

$$\mathcal{I}(x,y) = \int_0^\infty \frac{t^{1/4}\,\mathrm{d}t}{(x^2+t)(y^2+t)d(t)}, \qquad K(x,y) = c\,xy^{1/2}q_0(x)\overline{q_0(y)}\,\mathcal{I}(x,y).$$

We show below that $K \in L^2(\rho\otimes\rho)$. Granting this, take $f, h \in L^2(\rho)$ and $F = M^{1/2}f$. Tonelli's theorem and the Cauchy–Schwarz inequality give

$$\int_0^\infty \frac{t^{1/4}}{d(t)} \iint |g_t(x)g_t(y)f(y)h(x)|\,\mathrm{d}\rho(y)\,\mathrm{d}\rho(x)\,\mathrm{d}t = \frac{1}{c}\iint |K(x,y)F(y)h(x)|\,\mathrm{d}\rho\,\mathrm{d}\rho < \infty,$$

so $((I-Q_0)F, h) = \iint K(x,y)F(y)\overline{h(x)}\,\mathrm{d}\rho(y)\,\mathrm{d}\rho(x)$. Both $I - Q_0$ and the integral operator with kernel K are bounded. They agree on the dense set $\operatorname{Range} M^{1/2}$, hence everywhere. Thus $I - Q_0$ is Hilbert–Schmidt.

Lower bound for d. Let $\nu(t)$ be the number of l with $2^{-k_l} > \sqrt{t}$; then $\nu(t) = 0$ for $t \ge 1$, and ν is finite and nonincreasing. Every I_{k_l} with $l > \nu(t)$ lies in $(0, \sqrt{t}]$, where $t/(s^2+t) \ge 1/2$. Hence

$$d(t) \ge \frac{1}{2}\sum_{l>\nu(t)} m_l \ge \frac{1}{2\beta(\nu(t)+1)}.$$

Bound for $\mathcal{I}$. Let $a = \max(x,y)$, and use $t^{1/4}/(\min(x,y)^2+t) \le t^{-3/4}$. For $t \ge a^2$ we have $\nu(t) \le \nu(a^2)$ and $\int_{a^2}^\infty t^{-7/4}\,\mathrm{d}t = (4/3)a^{-3/2}$. For $t < a^2$, $\nu(t) - \nu(a^2)$ counts the k_l with $\sqrt{t} < 2^{-k_l} \le a$, so $\nu(t) \le \nu(a^2) + 1 + \log_2(a^2/t)$. Since $\int_0^1 u^{-3/4}(1 + \log_2(1/u))\,\mathrm{d}u < \infty$, we obtain, with a constant C depending only on β,

$$\mathcal{I}(x,y) \le 2\beta\int_0^\infty \frac{t^{-3/4}(1+\nu(t))}{a^2+t}\,\mathrm{d}t \le Ca^{-3/2}(1+\nu(a^2)).$$

Summation. Let $x \in I_{k_l}$, $y \in I_{k_{l'}}$ and $b = \min(x,y)$. The point a lies in $I_{k_{\min(l,l')}}$. Every l'' counted by $\nu(a^2)$ satisfies $2^{-k_{l''}} > a > 2^{-k_{\min(l,l')}-1}$, so $\nu(a^2) \le \min(l,l')$. Also $b/a < 2^{1-|k_l - k_{l'}|} \le 2^{1-|l-l'|}$. Since $x^2y/a^3 \le b/a$ in both cases $x = a$ and $y = a$,

$$|K(x,y)|^2\,\mathrm{d}\rho(x)\,\mathrm{d}\rho(y) = c^2x^2y\,\mathcal{I}(x,y)^2\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) \le c^2C^2\frac{b}{a}(1+\min(l,l'))^2\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y).$$

Integrating over $I_{k_l} \times I_{k_{l'}}$ and summing,

$$\|K\|^2_{L^2(\rho\otimes\rho)} \le 2c^2C^2\beta^2 \sum_{l\le l'} \frac{(1+l)^2}{l^2 l'^2} 2^{1-(l'-l)} \le 16c^2C^2\beta^2 \sum_{l'\ge 1} \frac{1}{l'^2} \sum_{r\ge 0} 2^{-r} < \infty. \qquad \square$$

Lemma 3. *Let q_0 satisfy (1) and (3), and suppose $q_0 \in \operatorname{Range} W^{1/2}$. Then $I - Q$ is not compact.*

Proof. The identity $(A^2x, x) = \|Wx\|^2 - |(Wx, q_0)|^2$ and (1) show that $A^2 \ge 0$. Equality would put Wx in the span of q_0, which (3) excludes for $x \ne 0$, so $\operatorname{Ker} A = \{0\}$. Taking adjoints in $QW^{1/2} = |A|^{1/2}$ gives $W^{1/2}Q^* = |A|^{1/2}$, so $\operatorname{Ker} Q^* = \{0\}$. Write $q_0 = W^{1/2}h$ with $h \ne 0$, and $C = Q^*Q$. By [2, Chapter 12, (10.5)], $CWC = W - (\cdot, W^{1/2}q_0)W^{1/2}q_0$ (that derivation uses only (10.3) and $\operatorname{Ker} W = \{0\}$). Hence

$$\|W^{1/2}Ch\|^2 = \|W^{1/2}h\|^2 - |(W^{1/2}h, q_0)|^2 = \|q_0\|^2 - \|q_0\|^4 = 0,$$

so $Ch = 0$ and $Qh = 0$. If $I - Q$ were compact, Q would be Fredholm of index zero, with closed dense range; then $\operatorname{Ker} Q^* = \{0\}$ would force $\operatorname{Ker} Q = \{0\}$, a contradiction. $\square$

Proof of Theorem 1. Let $\mu^\sharp = |q_0^\sharp|^2\rho$. It has the same null sets as ρ, so $\mu^\sharp(I_{k_l}) > 0$ for every l, and ρ is carried by $\bigcup_l I_{k_l}$. Given positive numbers m_l with $\sum_l m_l = 1$, define $\varphi = (m_l/\mu^\sharp(I_{k_l}))^{1/2}$ on I_{k_l} and $\varphi = 1$ elsewhere on $\sigma(W) \setminus \{0\}$. Then $q_0 = \varphi q_0^\sharp$ satisfies (1) and (2), and $|q_0|^2\rho(I_{k_l}) = m_l$.

For φ_{c} take $m_l = 1/(l(l+1))$; then $\sum_{l>n} m_l = 1/(n+1)$, so Lemma 2 applies with $\beta = 1$. It gives (3), and shows that $I - Q_0$ is Hilbert–Schmidt. By Lemma 1, $I - Q = (I - Q_0) \oplus 0$ is Hilbert–Schmidt.

For φ_{n} take $m_l = \kappa 2^{-k_l}/l^2$, with $\kappa > 0$ such that $\sum_l m_l = 1$. Then

$$\int \frac{|q_0|^2}{s}\,\mathrm{d}\rho \le \sum_l 2^{k_l+1} m_l = 2\kappa \sum_l \frac{1}{l^2} < \infty, \qquad \int \frac{|q_0|^2}{s^2}\,\mathrm{d}\rho \ge \sum_l 4^{k_l} m_l \ge \kappa \sum_l \frac{2^{l-1}}{l^2} = \infty.$$

So $q_0 \in \operatorname{Range} W^{1/2}$ and (3) holds; Lemma 3 shows that $I - Q$ is noncompact.

Finally fix $t > 0$. On $(t/2, 2t)$ the function φ takes finitely many positive values, so there it is bounded above and below. Off this interval $s^2/|s^2 - t^2| \le 4/3$, and the corresponding part of the integral in Equation (14.8) is finite for any $q_0 \in L^2(\rho)$. Hence the integral in Equation (14.8) is finite for q_0 exactly when it is finite for $q_0^\sharp$. $\square$

Remark 3. For $W = \operatorname{diag}(2^{-j})$, Lemma 2 with $\beta = \pi^2/6$ covers the explicit choice $q_0(j) = \sqrt{6}/(\pi j)$. Peller also notes that $I - Q$ is not Hilbert–Schmidt in general; Lemma 3 gives a noncompact choice for every Γ in [2, Chapter 12, Theorem 9.1] with trivial kernel.

Acknowledgements. AI tools assisted with the preparation of this manuscript.

References

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